

Aditya-L1 Solar Observatory Mission

Indian Space Research Organisation (ISRO) — 2023

1. Mission Overview

Aditya-L1 is India's first dedicated solar observatory mission. The spacecraft was launched on 2 September 2023 by the PSLV-C57 rocket from Satish Dhawan Space Centre, Sriharikota. The mission is designed to study the Sun from a halo orbit around the Sun-Earth Lagrange point 1 (L1), located approximately 1.5 million km from Earth. The L1 point provides a unique vantage point for continuous, uninterrupted observation of the Sun without any occultation or eclipse.

2. Scientific Objectives

The primary scientific objectives of Aditya-L1 are to study the solar corona and its heating mechanism, solar wind acceleration, coupling and dynamics of the solar atmosphere, solar wind distribution and temperature anisotropy, and the origin of coronal mass ejections (CMEs) and flares. Understanding these phenomena is crucial for space weather prediction, which affects satellite operations, power grids, and communication systems on Earth.

3. Scientific Payloads

Aditya-L1 carries seven scientific payloads with a total mass of approximately 244 kg. The payloads are: VELC (Visible Emission Line Coronagraph) — the primary payload, designed to study the solar corona and dynamics of coronal mass ejections, built by the Indian Institute of Astrophysics; SUIT (Solar Ultraviolet Imaging Telescope) for imaging the solar photosphere and chromosphere in near-UV; SoLEXS (Solar Low Energy X-ray Spectrometer) for monitoring solar X-ray flares; HELIOS (High Energy L1 Orbiting X-ray Spectrometer) for studying hard X-ray flares; ASPEX (Aditya Solar wind Particle EXperiment) for studying solar wind protons and alpha particles; PAPA (Plasma Analyser Package for Aditya) for measuring solar wind electrons and heavy ions; and MAG (Advanced Tri-axial High Resolution Digital Magnetometers) for measuring the interplanetary magnetic field.

4. The L1 Lagrange Point

The Sun-Earth Lagrange point 1 (L1) is a gravitational equilibrium point located approximately 1.5 million km from Earth in the direction of the Sun, which is about 1% of the Earth-Sun distance. A spacecraft at L1 requires minimal fuel to maintain its position. The key advantage of the L1 point for solar observation is that it allows continuous, uninterrupted viewing of the Sun without any occultation or eclipse caused by Earth or Moon. Aditya-L1 reached its halo orbit around L1 on 6 January 2024.

5. VELC Instrument

The Visible Emission Line Coronagraph (VELC) is the largest and most complex payload on Aditya-L1. It is designed to image the solar corona continuously and study the dynamics of coronal mass ejections (CMEs). VELC can simultaneously image the corona in three broadband channels and perform spectroscopy in one channel. It can observe the corona from 1.05 to 3 solar radii. The instrument is expected to transmit approximately 1440 images per day to the ground station.